

you need to have the bumper replaced. He goes to check their inventory levels and tells you he has a new bumper in stock and they can complete your repair immediately. This is good news for everyone—you because you get your car back sooner and the dealership because they have a happy customer and don't run the risk of your taking the car somewhere else for repair. Also, they don't have to store your car or give you a loaner while they wait for the part to come in.

In traditional mass production, the way to avoid stockouts—not having the product the customer wants—is to keep a large inventory of spares just in case. It may be that the blue 2011 Camry bumper is quite popular, but what about last year's model or the model from five years ago? The more inventory you keep, the greater the likelihood you will have the right product in stock for every customer. But large inventories are expensive because they have to be transported, stored, and tracked. What if the 2011 bumper turns out to have a defect? All the spares in all the warehouses instantly become waste.

Lean production solves the problem of stockouts with a technique called pull. When you bring a car into the dealership for repair, one blue 2011 Camry bumper gets used. This creates a “hole” in the dealer's inventory, which automatically causes a signal to be sent to a local restocking facility called the Toyota Parts Distribution Center (PDC). The PDC sends the dealer a new bumper, which creates another hole in inventory. This sends a similar signal to a regional warehouse called the Toyota Parts Redistribution Center (PRC), where all parts suppliers ship their products. That warehouse signals the factory where the bumpers are made to produce one more bumper, which is manufactured and shipped to the PRC.

The ideal goal is to achieve small batches all the way down to single-piece flow along the entire supply chain. Each step in the line pulls the parts it needs from the previous step. This is the famous Toyota just-in-time production method.⁸

When companies switch to this kind of production, their